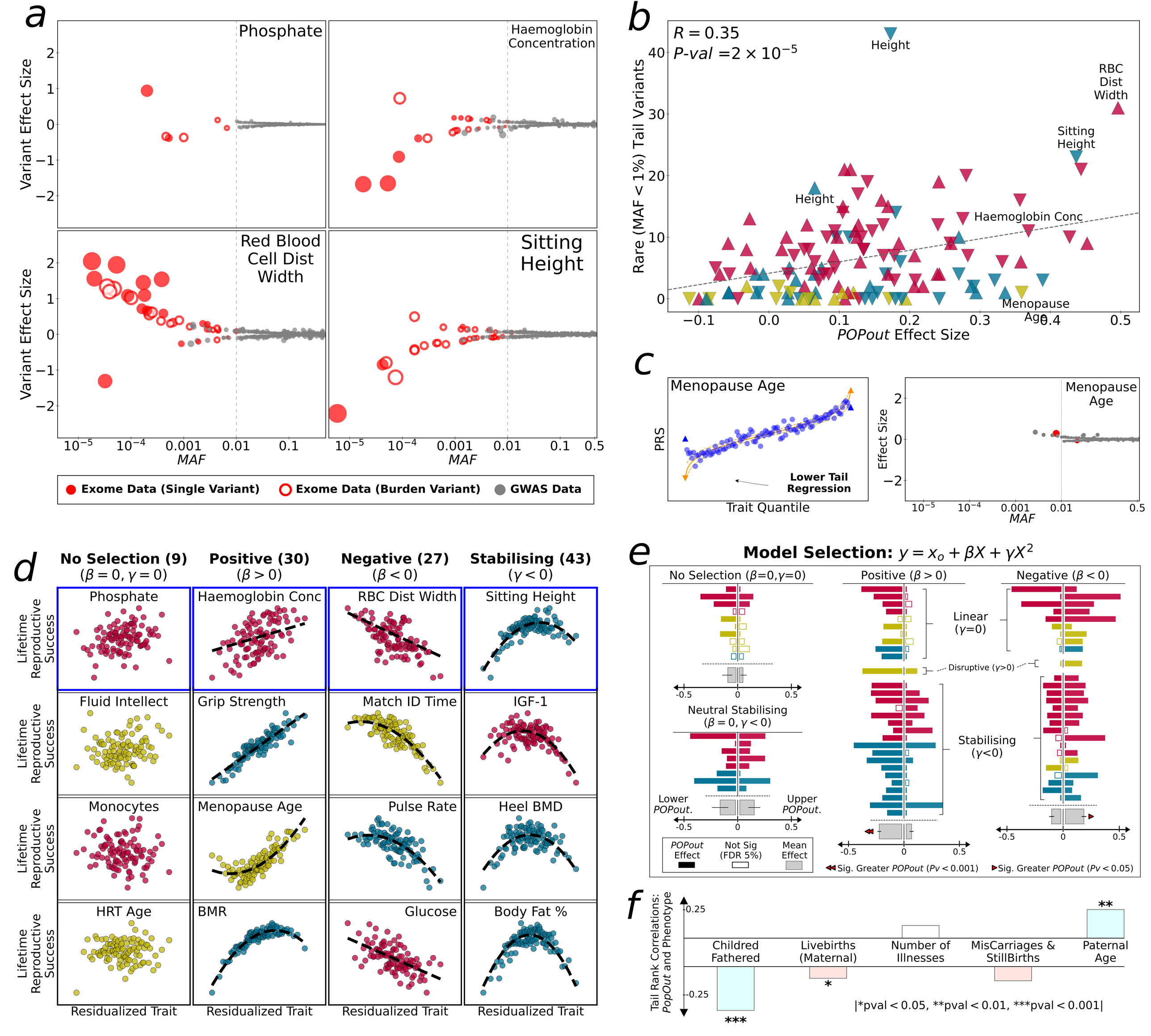
a
Phosphate
Haemoglobin Concentration
Red Blood Cell Dist Width
Sitting Height
Variant Effect Size
MAF
Exome Data (Single Variant)
Exome Data (Burden Variant)
GWAS Data
b
R = 0.35
P-val =2 × 10−5
Height
RBC Dist Width
Sitting Height
Height
Haemoglobin Conc
Menopause Age
Rare (MAF < 1%) Tail Variants
POPout Effect Size
c
Menopause Age
PRS
Lower Tail Regression
Trait Quantile
Menopause Age
Effect Size
MAF
d
No Selection (9) (β = 0, γ = 0)
Positive (30) (β > 0)
Negative (27) (β < 0)
Stabilising (43) (γ < 0)
Phosphate
Haemoglobin Conc
RBC Dist Width
Sitting Height
Fluid Intellect
Grip Strength
Match ID Time
IGF-1
Monocytes
Menopause Age
Pulse Rate
Heel BMD
HRT Age
BMR
Glucose
Body Fat %
Lifetime Reproductive Success
Residualized Trait
e
Model Selection: y = x_o + βX + γX²
No Selection (β=0,γ=0)
Positive (β > 0)
Negative (β < 0)
Linear (γ=0)
Disruptive (γ>0)
Neutral Stabilising (β = 0, γ < 0)
Stabilising (γ<0)
Lower POPout.
Upper POPout.
POPout Effect
Not Sig (FDR 5%)
Mean Effect
Sig. Greater POPout (Pv < 0.001)
Sig. Greater POPout (Pv < 0.05)
f
Tail Rank Correlations: PopOut and Phenotype
Childred Fathered
Livebirths (Maternal)
Number of Illnesses
MisCarriages & StillBirths
Paternal Age
|*pval < 0.05, **pval < 0.01, ***pval < 0.001|